

# Force and extension (RP)

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

| QUESTION |                                                                                                        | ANSWER                                                                  | ✓                        |
|----------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------|
| 1        | Investigating how force affects the extension of a spring: name the independent variable.              | Force applied to the spring                                             | <input type="checkbox"/> |
| 2        | Investigating how force affects the extension of a spring: name the dependent variable.                | Extension of the spring                                                 | <input type="checkbox"/> |
| 3        | State the extension of a spring when no force is applied.                                              | Zero                                                                    | <input type="checkbox"/> |
| 4        | Describe how to convert the mass hung on a spring into a weight.                                       | Multiply the mass in kg by the gravitational field strength (9.8 N/kg). | <input type="checkbox"/> |
| 5        | A student records the whole length of a spring, not its extension. Name the type of error this causes. | A systematic error (every value is too large).                          | <input type="checkbox"/> |
| 6        | The force on a spring is doubled. State the effect on its extension.                                   | It doubles (below the limit of proportionality).                        | <input type="checkbox"/> |